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1.0 SEM

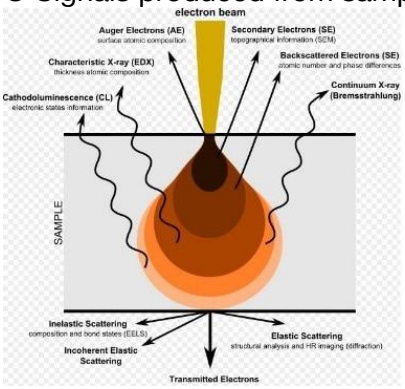
1.1 SEM_What is?

@ SEM, Scanning Electron Microscope.
@ used for observing the fine structure of a specimen surface at high magnification.
(TEM, Transmission Electron Microscope. Observing inner surface.)
@ Features of SEM:

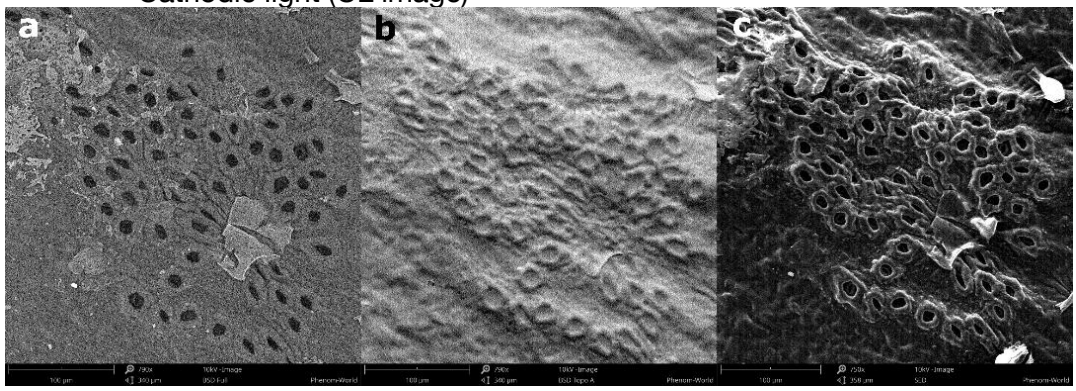
- All solid surface, from low to high magnification
- Greater Focal depth (higher aperture), stereoscopic(3D) image.

1.2 SEM_Why is used?

@ Scan a sample surface, with finely converged electron beam in vacuum.
@ Signals produced from sample:



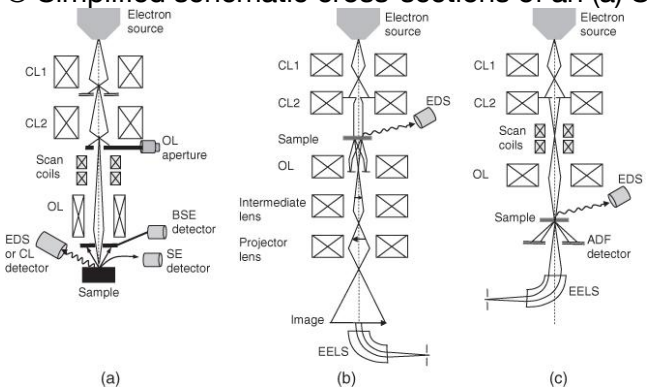
- Incident electron beam
- Secondary electrons (SE image) – topographical structure of the sample
- Backscattered electrons (BSE image) – shows high sensitivity to differences in atomic number, the higher the atomic number, the brighter the material appears in the image; Provide crystallography, topography, and magnetic field of the sample.
- Characteristic X-rays
- Cathodic light (CL image)

@ 

a) Full backscatter electron detector (BSD), b) topography BSD and c) secondary electron detector images of a leaf.

1.3 SEM_How it work.

@ Simplified schematic cross-sections of an (a) SEM, (b) TEM and (c) STEM instrument.



@ -Electron Gun (3 type: Field emission electron gun, Schottky electron gun, thermal electron gun)

-Condenser lens – electromagnetic coils, converge the electron beam into a fine beam.

-Deflection coil-X& Y direction and change the area(mag) to be scan.

-Objective Lens- converge into fine beam; 3 types of lens, in-lens, semi in lens, and out-lens.

-Secondary electron detector- convert electric signal and amplifies the SE produced.

-Display

-Vacuum pump -Evacuates the column and specimen chamber

1.4 SEM_Procedure for SEM Observation.

@1. Startup of instrument
@2. Preparation of sample to be observed
@3. Setting of sample to microscope
@4. Observation of sample
@5. Acquisition and saving of image

1.5 SEM_Sampling

@ Tools Used

-Specimen stub, Double sided Carbon Tape, Conducting Graphite Paint, Tweezers, Wafer Tweezers, Blower, Diamond Scribing Pen.

@ Sampling

- 1. Bulk sample, film, etc.
- 2. Powder sample
    - Sprinkling Method: Apply it on to carbon paste and blow it.
    - Suspension method: Sample +non-chemical solution >> disperse it using ultrasonic cleaner, Apply drops of the suspension onto aluminium foil, and spread it out using a blower or the like, cut and put it to specimen stub.
- 3. Wafer or glass sample.
    - trimming sample, using diamond scribing pen and ruler.
- 4. Biological sample

- Perfusion fixation (Substitution of fixation solution with blood and glutaraldehyde, formalin. etc)
- Fine sectioning od sample
- Immersion fixation (Glutaraldehyde, formalin, etc)
- Post-fixation (Osmium tetroxide)
- Conductive staining (Tannic acid – osmium tetroxide, etc.)
-

- 5. Foodstuffs, samples containing oil
  - these are observed with a low vacuum SEM.

@ Metal Coating

- 1. Purpose of coating
  - to make the sample surface conductive (prevent of charge-up)
  - to increase the yield of secondary electrons (increase signal)
  - to prevent damage sample
- -2. Film thickness of coating
  - thickness of nanometers is desirable.
  - High magnification, thin coating.
  - Low magnification, thick coating.
- 3. Kinds of metal targets
  - Coating particles >> Fine
  - Au >> Au-Pd >> Pt-Pd >> Pt
  - Film, adequately fine particles.
  - Enough to emit secondary electron.
  - Stable against oxidation.
- 4. C (carbon) coating
  - vacuum evaporator
  - suitable for elemental analysis

@ Ion Milling

- -1. What is ion milling
  - A technique that utilize a **board ion beam (BIB) discharged** from an ion gun to sputter atoms at the sample surface.
  - used for **eliminating machining flaws** from the sample surface in preparing cross-sectional samples of multiple film layers.
- -2. What is BIB ion milling?
  - Surface processing (Flat milling method)
    - # sample surface is uniformly etched by irradiating the ion beam at an angle while rotating the sample.
    - # Elimination of oxide film or contamination.
    - # Elimination of machining flaws
    - # Observation of crystal grains or layers formed by relief milling.

  Cross-section processing
    - # Observation of cracks or voids which is difficult to be prepared by mechanical polish process.
    - # Overservation/Analysis of interface of multilayered specimen.
    - # Cross-sectional observation of sample (paper/film, etc.) susceptible to stress.

1.6 SEM_Observe with SEM

@@ Metals, electronic materials
@ Large Sample
@ Insulating material
@ Interior, cross-section sample
@@ Polymeric materials
@ Sample susceptible to heat
Eg: Membrane filter (low vacuum mode/ ultra-low acceleration voltage,100v)
@ Insulating material
Eg: Rubber (low vacuum mode)
@ Liquid, emulsion
Eg: Latex (freeze with nitrogen gas, then fractured surface is observed, FESEM + Cryogenic system)

@Decelation mode:
- -ve voltage is applied to the sample, and the electron beam accelerated via the electron gun is decelerated just before the sample.
- observe sample sample surface at low irradiating voltages of 500V or less that are difficult with usual SEM.

@Cryogenic system
- Rapidly freese a sample containing moisture or oily component as it is for observation with the SEM.

@EBSP (Electron Backscattering Pattern)
- method of electron beam backscattering diffraction, and by attaching the system (EBSD) that analyzes this to the SEM.
- the crystal orientation of submicron order of a sample can be identified and mapping images of the crystal orientation can be acquired.

@@Power, micro-particles, nano materials
@Samples using the sprinkling method, double sided carbon tape

@@Toner, Cosmetics
@Toner
@Pigment

@@@Semiconductor Wafer, Liquid Crystal Display
@@ Semiconductor
@ Surface observation, Cross-section observation, Observation of thin film

@@ Liquid Crystal
@ Observation of transparent electrode surface
@ Observation of TFT (thin-film transistor) cross-section.

@@@ Biological Samples
@ Insects, @Plant
@ Aquatic organisms
@ Tissue, cells (fixation-dehydration-drying)
@ Microorganisms (fixation-dehydration-drying)

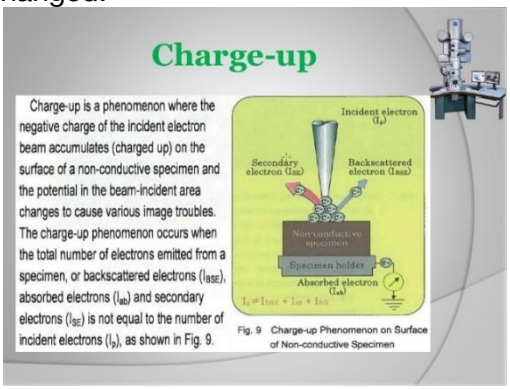
@@@ Foodstuffs
@ Food constitution, processed food.
@ Sample susceptible to heat
@ Sample with high moisture content

1.7 SEM_Causes of these image

A. Charge Up	B. Contamination	C. Beam damage	D. Effect of external disturbance	E. Other

	A	B	C	D	E
Image Moves	■		■	■	
Sample Structure cannot be discerned (recognize)	■	■	■		
Image is distorted (twisted out of shape)	■	■	■	■	■
Image fluctuates (rise and fall irregularly)	■				■
Brightness is unstable	■	■			■
Cannot be focused	■	■	■		■

A. Charge Up
- Observation of non-conductive sample.
- Conspicuous (clearly visible) especially when scan speed or magnification is changed.



-
- Countermeasures for the charge-up phenomena:
    1. Reduce accelerating voltage.
    2. Reduce the sample irradiating current.
    3. Apply a metal coating.
    4. Integrate the image (form an image by superimposing images obtained at rapid scan)
    5. Observe image in low vacuum mode.
    6. Utilize a low-acceleration BSE signal (eliminate SE signal by means of signal varying mechanism).

B. Contamination
- gas molecules of hydrocarbons existing around the sample collect on the sample due to electron beam irradiation.
- To reduce contamination:

Reduction of residual gas molecules (improvement of vacuum level)
Reduction of gas molecules derived from sample

- To achieve reduction:
  # Use min of conductive paste.
  # dry the conductive paste with dryer
  # Heat and degas the sample in vacuum device.
  # focus and observe a quickly.
  # Observe samples while cooling the sample surroundings with a cold trap.

C. Bean Damage
- Thermal change of chemical change occurring due to electron beam irradiation.
- Temperature rise of the sample due to electron beam is dependent on a number of factors including acceleration voltage and intensity of beam, observation area, observation time, specific heat, heat conductivity and others.
- Eg: polymeric materials and/or biological samples.
- Countermeasure beam damage:
  1. Reduce the sample irradiating current.
  2. Lower the accelerating voltage.
  3. Appl metal coating to the sample (improve heat conductivity)
  4. Observe the sample while cooling it.

D. Effect of disturbance
- Fringes/distortion causes by vibration or a stray magnetic field.
- Countermeasure of due to vibration.
  1. Keep the instrument away from the vibration sources.
  2. do not let high-voltage cables from the column come in contact with the wall or other installation items.
  3. No draft from an air conditioner outlet contact the column directly.
- Countermeasure of due to magnetic field.
  1. keep away from the magnetic field source, such as cable.
  2. Shorten the working distance, and apply strong excitation to the condenser lens to counter the effect of a magnetic field.
  3. use a magnetic field cancellation.

E. Other causes of image abnormalities

Symptom	**Possible Causes**
Sample moves	- Sample not fixed in place - Specimen holder not tightened - Sample is inserted incompletely - Compressor operated while stage is locked.
Image fluctuates	- Irradiating current is low (change the excitation of condenser lens) - Lower image is being observed at short WD in the case of semi in lens SEM. - WD is long in low vacuum SEM observation
Focus cannot be obtained	- Inadequate (lacking the quality) optical axis alignment. - Objective aperture contaminated. - Recheck the instrument parameter.

1.8 SEM_SEM instruments available

1. **SEM & VP-SEM:** Standard and Variable-Pressure Scanning Electron Microscopes,
2. **FE-SEM:** Field-Emission Scanning Electron Microscopes,
3. **TEM:** Biological and Analytical Transmission Electron Microscopes (TEM),
4. **STEM:** Scanning Transmission Electron Microscopes (STEM), and
5. Tabletop Microscopes.



Ultra-high Resolution Scanning Electron Microscope SU9000

The Cold Field Emission source is ideal for high-resolution imaging with a small source size and energy spread. Innovative CFE Gun technology contributes the ultimate FE-SEM with superior beam brightness and stability, affording high-resolution imaging and high-quality elemental analysis. Unique object lens design has a capability of EELS and diffraction as well.

Ultra-high Resolution Scanning Electron Microscope Regulus Series

As a new brand of FE-SEMs, the Regulus series lineup comprises four models: the Regulus8100, Regulus8220, Regulus8230, and Regulus8240, all of which extend the functions of the SU8200 series with the use of a common platform.

Ultra-High-Resolution Schottky Scanning Electron Microscope SU7000

The SU7000 is designed to allow simultaneous acquisition of multiple secondary and backscattered electron signals, and enables rapid capture of many types of signals. With the ability to display and store up to 6 signal channels simultaneously, the SU7000 offers unsurpassed imaging performance. In addition, it provides a flexible of specimen chamber and vacuum system to support the broad range of observational conditions. Moreover, the electron gun—with its built-in Schottky emitter—can provide irradiating beam currents of up to 200 nA. The SU7000 is built to accommodate the full diversification of future analytical methods.

Schottky Field Emission Scanning Electron Microscope SU5000

Innovative analytical FE-SEM allows for a simple transition between high vacuum and variable pressure mode. EM Wizard is a knowledge-based system for SEM imaging that goes beyond basic preset conditions and recipes. Its ease of use opens a new gateway for material research, development, and area beyond our imagination.



MirrorCLEM System for Correlative Light and Electron Microscopy

MirrorCLEM, a simple solution for correlative light and electron microscopy (CLEM).



Scanning Electron Microscope SU3800/SU3900

Performance & Power in a Flexible Platform Hitachi High-Tech's scanning electron microscopes SU3800/SU3900 deliver both operability and expandability. The operator can automate many operations and efficiently utilize their high performance. The SU3900 is equipped with a large multipurpose specimen chamber to accommodate observation of large samples.



Scanning Electron Microscope FlexSEM 1000 II

The FlexSEM 1000 II VP-SEM combines innovative technological features with an intuitive interface, to deliver adaptability and flexibility in a powerful, automated, lab-friendly package. Cutting-edge technology and circuitry provides unrivaled imaging performance, even in variable-pressure environments, a feature previously only available in a full-sized SEM. This SEM runs on clean energy for an economical analytical tool, without compromising performance. The FlexSEM will change your view of electron microscopy!



Tabletop Microscopes TM4000II / TM4000Plus II

Newly released TM4000 II series provide 20kV accelerating voltage and also allow wide area of SEM observation by adding Multi Zigzag function (Option) Tabletop Micro scope offers further enhanced observation and analysis application world.



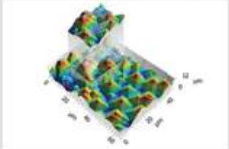
TM Series Energy Dispersive X-Ray Spectrometers: AZtec Series

TM Series Energy Dispersive X-Ray Spectrometers: AZtec Series. The TM4000 Series offers multiple EDS systems to choose from based on application and budget. Two classes of detectors are available: a low-cost 10-mm2 detector and a large-sized 30-mm2 detector, both of which are of compact design and do not require LN2.



TM Series Energy Dispersive X-ray Spectrometer: Quantax75

TM Series Energy Dispersive X-ray Spectrometer: Quantax75. Hitachi Tabletop microscope TM4000 Series offers multiple EDS systems to choose from based on application and budget. All detectors offered are of compact design and do not require LN2.



The TM Series 3D visualization software Hitachi map 3D

Hitachi map 3D: Three-dimensional models allow height measurements. A 3-dimensional model can be generated without sample tilting and alignment, using 4-segment backscattered electron detector.



Field Emission Transmission Electron Microscope HF5000

Hitachi's unique 200 kV aberration-corrected TEM/STEM: the perfect harmony of imaging resolution and analytical performance 0.078 nm spatial resolution in STEM is achieved together with high specimen-tilt capability and large solid angle EDX detector(s), all in a single objective lens configuration. The HF5000 builds on features from Hitachi HD-2700 dedicated STEM including Hitachi's own fully automated aberration corrector, symmetrical dual SDD EDX and Cs-corrected SE imaging. It also incorporates the advanced TEM/STEM technologies developed in the HF series. Integrating these accumulated technologies into a new 200 kV TEM/STEM platform results in an instrument with an optimum combination of sub-Å imaging and analysis, as well as the flexibility and unique capabilities to address the most advanced studies.



Transmission Electron Microscope H-9500

The H-9500 is a 100-300 kV TEM with an LaB6 electron gun. This is a user-friendly workhorse for atomic-resolution TEM imaging and routine structural characterization. The excellent imaging capability also makes the H-9500 a platform for in-situ TEM. Various Hitachi in-situ heating TEM specimen holders* enable the imaging of dynamic structural changes at elevated temperatures. A special version for gas environmental in-situ TEM is available.



Transmission Electron Microscope HT7800 Series

The HT7800 RuliTEM is a 120 kV transmission electron microscope (TEM) with multiple lens configurations, including a standard lens for unsurpassed high contrast and a class-leading HR lens for high resolution.

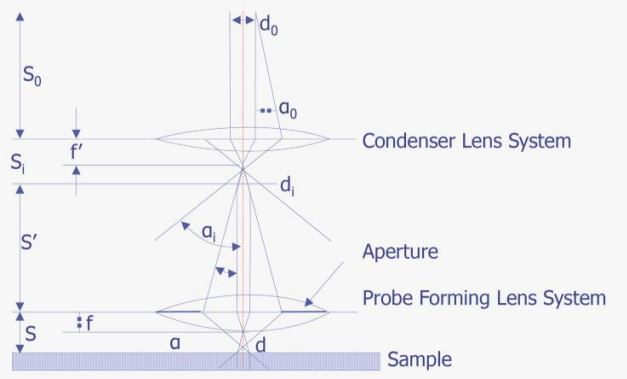


Advanced Mineral Identification and Characterization System (AMICS)

The Advanced Mineral Identification and Characterization System (AMICS) is the latest software package for automated identification and quantification of minerals and synthetic phases. The core of this software package is its innovative imaging and analysis capabilities. It enables Bruker's QUANTAX energy dispersive X-ray spectrometry (EDS) systems on Hitachi scanning electron microscopes (SEM) to become a fully automated Mineral Liberation Analyzer (MLA).

1.9 SEM_Theory

1.9.1 Formation of electron beam.



1.9.1.1 How is a fine electron beam formed?

- virtual electron source
- diameter d0 → d
- condenser and objective lenses of SEM are both electromagnetic lenses.
- magnification M1, and M2.
- M1 = Si/S0; M2 = S/S'
- d = d0*M1*M2
- Limitation:
- X, reduce d by lower M1, M2
- when M1 lowered (condenser lens current increase), electrons incident on the sample will also be reduced.
- the brighter the electron gun, the smaller d can.
- the thickness of the electron beam never becomes smaller than the diffraction spread.
- dd, diffraction limit is proportional to λ (the e's wavelength), λ is inversely proportional to the square root of the accelerating voltage (√Vacc). When using a bright field emission gun determines whether

1.9.2 Evacuation
1.9.3 Generation, detection and usage of SEM signals
1.9.4 Viewing conditions for acquiring good SEM images.
1.9.5 Principle and applications low vacuum SEM
1.9.6 Principle and applications of STEM
1.9.7 How to generate and detect x-rays and perform elemental analysis
1.9.8 How to improve precision in X-ray analysis
1.9.9 Concerning others analysis equipment (EBSP, CL)

1.4 SEM_Different Brand/ Focus Point

1.5 SEM_Technical Part

1.6 SEM_Compare with other similar test

1.7 SEM_Limitation

1.8 SEM_References/Book

1. Low Voltage Electron Microscope Principles and Applications; David C. Bell; WILEY
2.
3.
4.

2.0 EDX

2.1 EDX_What is?

Energy-dispersive X-ray spectroscopy (EDS, EDX, EDXS or XEDS)
Energy dispersive X-ray analysis (EDXA) / Energy dispersive X-ray microanalysis (EDXMA).

2.2 EDX_Why is used?

- Elemental analysis or chemical characteristics.

2.3 EDX_How it work.

- It relies on an interaction of some source of X-ray excitation and a sample.
-To stimulate the emission of characteristic X-rays from a specimen a beam of X-rays is focused into the sample being studied. At rest, an atom within the sample contains ground state (or unexcited) electrons in discrete energy levels or electron shells bound to the nucleus. **The incident beam may excite an electron in an inner shell**, ejecting it from the shell while creating an electron hole where the electron was. **An electron from an outer, higher-energy shell then fills the hole**, and the difference in energy between **the higher-energy shell and the lower energy shell may be released in the form of an X-ray**. The number and energy of the X-rays emitted from a specimen can be measured by an energy-dispersive spectrometer. As the energies of the X-rays are characteristic of the difference in energy between the two shells and of the atomic structure of the emitting element, EDS allows the elemental composition of the specimen to be measured.

2.4 EDX_Detector

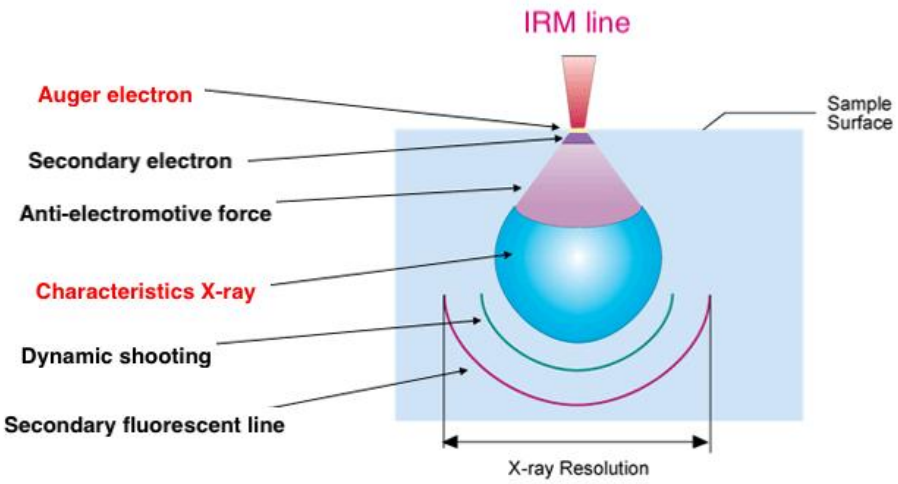
The most common detector used to be **Si(Li) detector** cooled to cryogenic temperatures with liquid **nitrogen**. Now, newer systems are often equipped with **silicon drift detectors (SDD)** with Peltier cooling systems.

2.4 EDX_Different Brand/Focus Point.

2.5 EDX_Technical Part.

2.6 EDX_Compare with other similar test.

EDX vs Auger (AES)
https://www.eag.com/resources/appnotes/what-is-the-relationship-between-auger-analysis-and-eds-analysis/#:~:text=In%20Auger%20analysis%20for%2C%20an,energy%20of%20the%20electron%20beam.



*IRM=Interference reflection microscopy
Auger analysis for, an information depth of about 10 nm or less can be obtained from a small spot irradiated with the incident electron beam.
EDS analysis, the characteristic X-rays come from a much large volume, that depends on the acceleration energy of the electron beam. The information comes from a much larger volume (about 1um3) than the irradiated spot that the X-ray actually hits. (In normal EDS analysis, particles with a size of 0.5um or more are targeted.)
This can be an advantage for EDS in some cases (particle/layer/feature sizes of **0.5um or larger**) and an advantage for Auger in other cases (smaller features and thin films) Analysis of a particle with a size of about 100nm..

2.7 EDX_Dectection Area.

2.8 EDX_Best Analysis Energy

EDX- the

3.0 FTIR

https://www.thermofisher.com/sg/en/home/industrial/spectroscopy-elemental-isotope-analysis/spectroscopy-elemental-isotope-analysis-learning-center/molecular-spectroscopy-information/ftir-information.html

FTIR subtopics



FTIR Basics

Your starting point for learning about FTIR, this section covers what "FTIR" stands for and how it works. We also introduce the main sampling techniques and provide a more extensive overview of how they can be used to answer your questions.



FTIR Sampling Techniques

An in-depth training section, this will cover transmission, reflection, DRIFTS and ATR accessories, their use and care. The reasons for choosing one or the other method will become clear, and the strengths of each method will be highlighted. Stress here is on sample preparation and experiment optimization.



FTIR Applications

This area contains a lot of content, from webinars and videos to application notes. A wide breadth of applications in FTIR will be found, basic to advanced, and more technical content covering instrument performance and operations. You will find considerable content about FTIR Microscopy in this section as well.



FAQs

Our market leading expertise in FTIR and FTIR Microscopy is not just about the instruments, but it is due to our wide experience and knowledgeable staff. We have gathered here many of the questions we have been asked, and will add to this section frequently.

3.1 FTIR_What is?
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4.0 EBSD

4.1 EBSD_What is?
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5.0 X-sect

5.1 X-sect BGA Ball

-IMC thickness spec
https://mpe.researchmfg.com/imc-soldering-joints/
There are many people keep chasing working-bear what is the ideal thickness of IMC layer? I will say the best thickness of IMC layer shall be 1~3um but the 1~5um is the regular acceptable thickness for the intermetallic compound of Copper-Tin (CuSn) and Copper-Nickel (CuNi).

As far as I know there is no standard thickness minimum or maximum thickness. It depends

-IMC thin cause crack
https://mpe.researchmfg.com/imc-soldering-joints/

XX.2 I am a Manufacturing Process Engineer (MPE)
XX.3 @@_How it work.
XX.4 @@_Different Brand/ Focus Point.
XX.5 @@_Technical Part.
XX.6 @@_Compare with other similar test.

XX.0 @@

XX.1 @@_What is?
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V·T·E	**Spectroscopy** [hide]
Vibrational	FT-IR · Raman · Resonance Raman · Rotational · Rotational–vibrational · Vibrational · Vibrational circular dichroism
UV–Vis–NIR	Ultraviolet–visible · Fluorescence · Vibronic · Near-infrared · Resonance-enhanced multiphoton ionization (REMPI) · Raman optical activity spectroscopy · Raman spectroscopy · Laser-induced
X-ray and photoelectron	Energy-dispersive X-ray spectroscopy · Photoelectron · Atomic · Emission · X-ray photoelectron spectroscopy · EXAFS
Nucleon	Gamma · Mössbauer
Radiowave	NMR · Terahertz · ESR/EPR · Ferromagnetic resonance
Others	Acoustic resonance spectroscopy · **Auger spectroscopy** · Astronomical spectroscopy · Cavity ring-down spectroscopy · Circular dichroism spectroscopy · Coherent anti-Stokes Raman spectroscopy · Cold vapour atomic fluorescence spectroscopy · Conversion electron Mössbauer spectroscopy · Correlation spectroscopy · Deep-level transient spectroscopy · Dual-polarization interferometry · Electron phenomenological spectroscopy · EPR spectroscopy · Force spectroscopy · Fourier-transform spectroscopy · Glow-discharge optical emission spectroscopy · Hadron spectroscopy · Hyperspectral imaging · Inelastic electron tunneling spectroscopy · Inelastic neutron scattering · Laser-induced breakdown spectroscopy · Mössbauer spectroscopy · Neutron spin echo · Photoacoustic spectroscopy · Photoemission spectroscopy · Photothermal spectroscopy · Pump–probe spectroscopy · Saturated spectroscopy · Scanning tunneling spectroscopy · Spectrophotometry · Time-resolved spectroscopy · Time-stretch · Thermal infrared spectroscopy · Video spectroscopy · Vibrational spectroscopy of linear molecules

Software development

Core activities

Processes · Requirements · Design · Engineering · Construction · Testing · Debugging · Deployment · Maintenance

Paradigms and models

Agile · Cleanroom · Incremental · Prototyping · Spiral · V model · **Waterfall**

Methodologies and frameworks

ASD · DevOps · DAD · DSDM · FDD · IID · Kanban · Lean SD · LeSS · MDD · MSF · PSP · RAD · RUP · SAFe · Scrum · SEMAT · TSP · OpenUP · UP · XP

Supporting disciplines

Configuration management · Documentation · Software quality assurance (SQA) · Project management · User experience

Practices

ATDD · BDD · CCO · CI · CD · DDD · PP · SBE · Stand-up · TDD

Tools

Compiler · Debugger · Profiler · GUI designer · Modeling · IDE · Build automation · Release automation · Infrastructure as code · Testing

Standards and Bodies of Knowledge

BABOK · CMMI · IEEE standards · ISO 9001 · ISO/IEC standards · PMBOK · SWEBOK · ITIL · IREB

Glossaries

Artificial intelligence · Computer science · Electrical and electronics engineering

Outlines

Outline of software development

V · T · E